IAS PRELIMS/MAINS

Ecology & Environment

A COMPREHENSIVE COVERAGE

MODULE - 34

SOLID WASTE - 4

CHAPTER - I – SOLID WASTE

E - WASTE

- It refers to the discarded electronic equipment.
- E-waste is not hazardous if it is stored safely or recycled by scientific methods.
- E-waste is hazardous if it discarded and disposed off using primitive methods.

HARMFUL SUBSTANCES IN E-WASTE

- **Lead** – Causes damage to the central and peripheral nervous system.
- **Mercury** – It is estimated that 22% the yearly world consumption of mercury is used in electrical and electronic equipment. Mercury causes brain and kidney damage.
- **Chromium** – It cause damage to DNA and is extremely toxic.
- **Brominated Flame Retardants** - These are used in circuit form retardants (BFRs) to reduce flammability.
- **Barium** – Short term exposure to barium causes brain swelling, muscles weakness, damage to heart, liver and spleen.
- **Beryllium** – Causes lung cancer and skin diseases.
- **Phosphor** - carcinogen
- And many other harmful substances.

Learning Space

E – WASTE IN INDIA

- India generates about 2 million tons of e-waste annually and is the 5th biggest generator of e-waste in the world, after US, China, Japan and Germany.
- The increased quantity of E-Waste is because of increased consumption and also due to people discarding old electronic devices much faster.
- According to a study the volume of E-Waste is growing at a rate of 20% annually.
- Unlike developed countries which have technology, developing countries use manual labour for dealing with e-waste, thus exposing workers to harmful substances.

E – WASTE IN INDIA

- Only 2-3% of India's total E-waste gets recycled causing depletion of natural resources and irreparable damage to environment and health of the people working in the industry.
- Over 95% of the E-waste generated is managed by the unorganized sector and scrap dealers who dismantle and dispose it to extract metals.
- E-waste accounts for most of the heavy toxic metals found in landfill . These lead to ground water pollution and soil acidification.

Learning Space

E – WASTE MANAGEMENT RULES - 2016

1 EXTENDED PRODUCER RESPONSIBILITY (EPR)

- Under these producers of electrical and electronic equipment have mandatory targets to collect e-waste and they should ensure that the e-waste is channelized to authorized recyclers.
- Under EPR the company must take back a certain proportion of their sales of a particular year.
  - ✓ Ex: A smart phone producing company has to take back a certain proportion of the phones it sold 5 years back, the age of a smartphone is estimated to be 5 years.
- The producers are liable to collect 10 per cent of e-waste during 2017-18, 20 per cent during 2018-19, 30 per cent during 2019-20 and 40 per cent during 2020-21 and so on.

E – WASTE MANAGEMENT RULES - 2016

1 EXTENDED PRODUCER RESPONSIBILITY (EPR)

- Extended producer responsibility is now applied to Retailers , Dealers and Manufacturers.
- Manufacturer has to collect E-waste during manufacturing of Electrical and Electronics components and safely recycle them.
- Collection based approach has been adopted to include collection centers and collection points for collection of E-waste by producers under EPR.
- Provision for Pan-India EPR authorization by CPCB instead state wise EPR authorization.

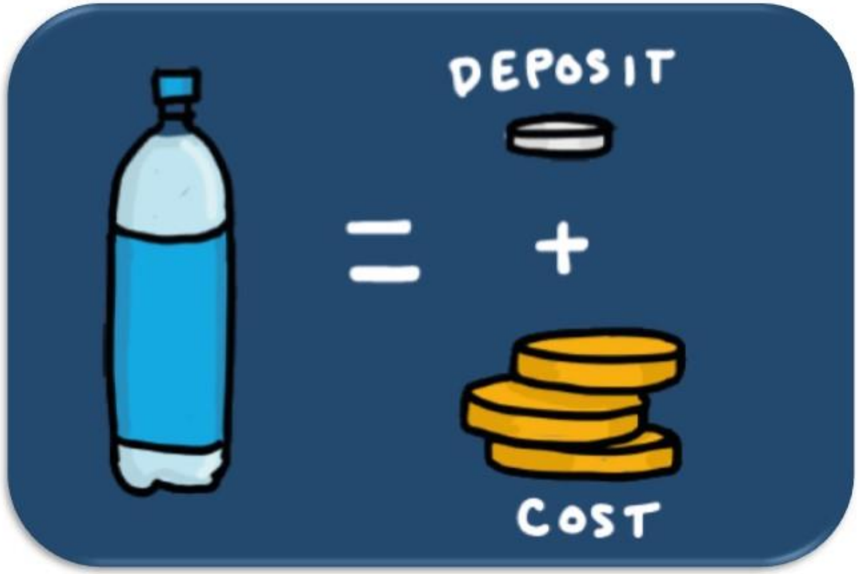
E – WASTE MANAGEMENT RULES - 2016

2 WIDER APPLICABILITY OF THE RULES

- Extended to components and spare parts of Electrical and Electronic Equipment (EEE), because bulk of E-waste comprise of spares and parts of EEE.
- Compact fluorescent lamp (CFL) and other mercury containing lamps brought under the purview of the rules.

3 MEASURES OF STREAMLINING THE COLLECTION OF E-WASTE

Option has been given for producers to introduce schemes such as deposit refund scheme.



E – WASTE MANAGEMENT RULES - 2016

4 THE ROLE OF THE STATE GOVT

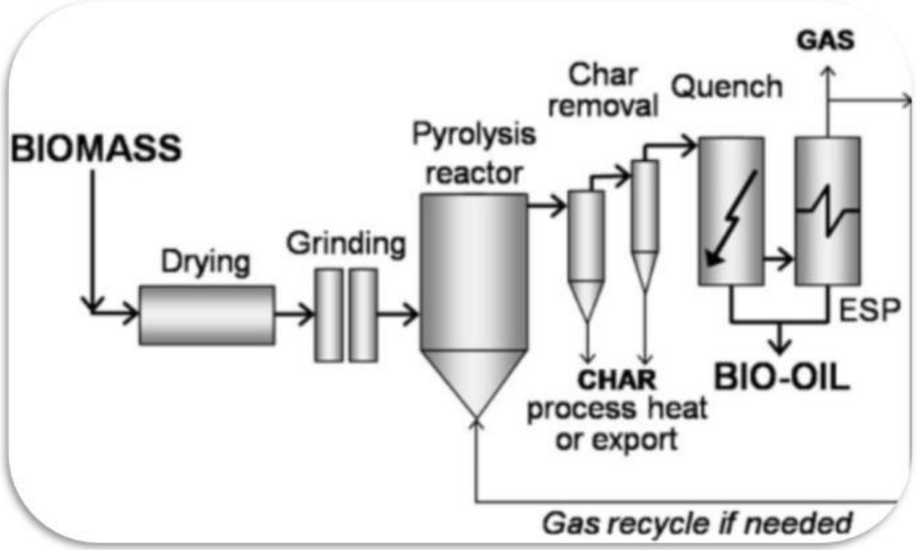
- The role of the state govt. also has been specified, which is
  a) The Department of Labor has to ensure safety, health and skill development of workers involved in E-waste sector.
  b) Department of Industry in the state must ensure earmarking of industrial space for E-waste dismantling and recycling facilities in the existing industrial estates and upcoming industrial estates.
  c) Department Labor should ensure registration and recognition of people involved in this work.

सत्यमेव जयते
MINISTRY OF LABOUR AND EMPLOYMENT
GOVERNMENT OF INDIA

सत्यमेव जयते
Government of India
Ministry of Commerce & Industry
Department of Commerce

PYROLYSIS

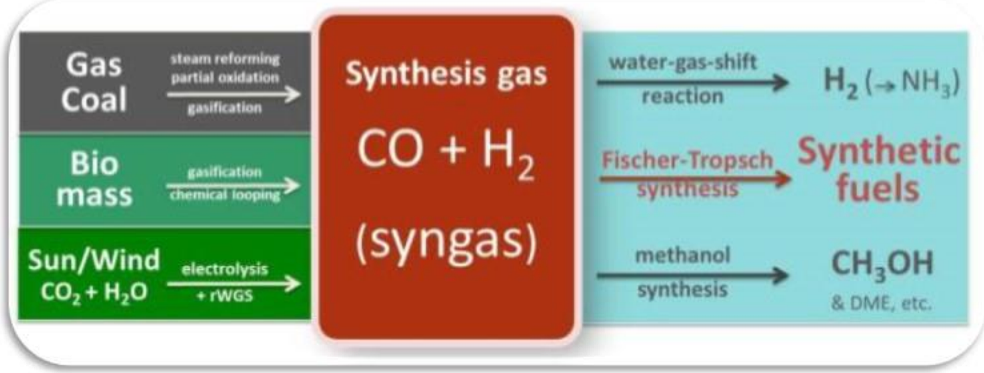
- Pyrolysis is the heating of an organic material, such as biomass, in the absence of oxygen. Because no oxygen is present the material does not combust but the chemical compounds make up that material thermally decompose into combustible gases and charcoal.
- Most of these combustible gases can be condensed into a combustible liquid, called pyrolysis oil (bio-oil).
- So, pyrolysis of biomass produces three products - one liquid, bio-oil, one solid, bio-char and one gaseous (syngas).

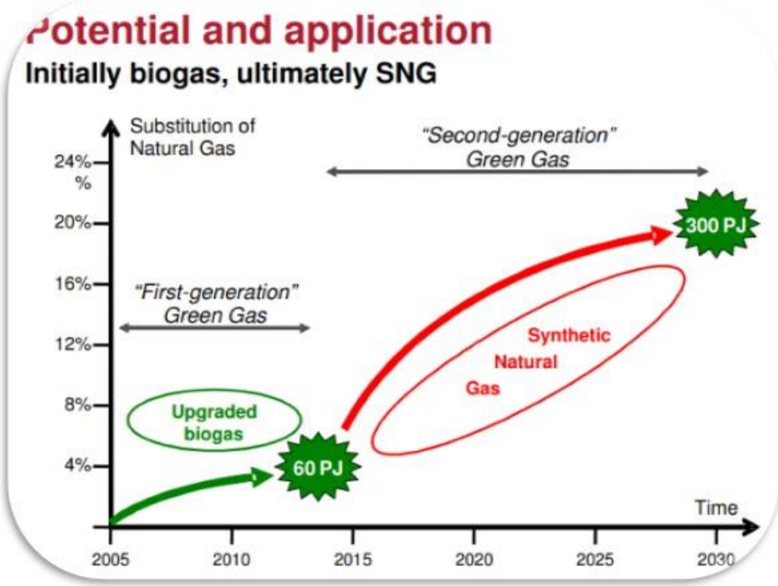


Learning Space

PYROLYSIS

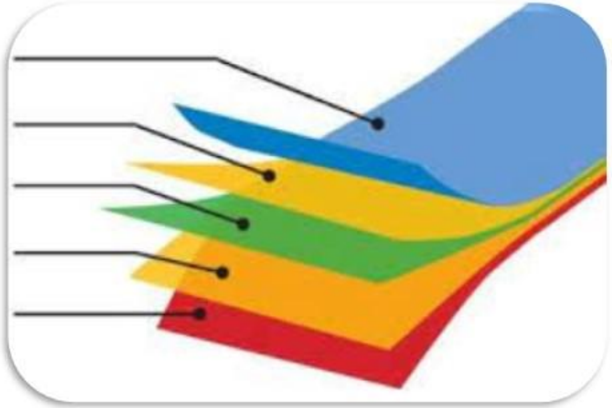
- Syngas, or synthesis gas, is a fuel gas mixture consisting primarily of hydrogen, carbon monoxide, and some carbon dioxide. The name comes from its use as an intermediate in creating Synthetic Natural Gas (SNG) and for producing ammonia or methanol.
- The pyrolysis process can be self-sustained, as combustion of the syngas and a portion of bio-oil or bio-char can provide all the necessary energy to drive the reaction.





PLASTICS THAT CAN BE CONVERTED INTO FUEL USING PYROLYSIS

- Mixed plastic (HDPE, LDPE, PE, PP, Nylon, Teflon, PS, FRP etc.)
- Multi-layered plastic.



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